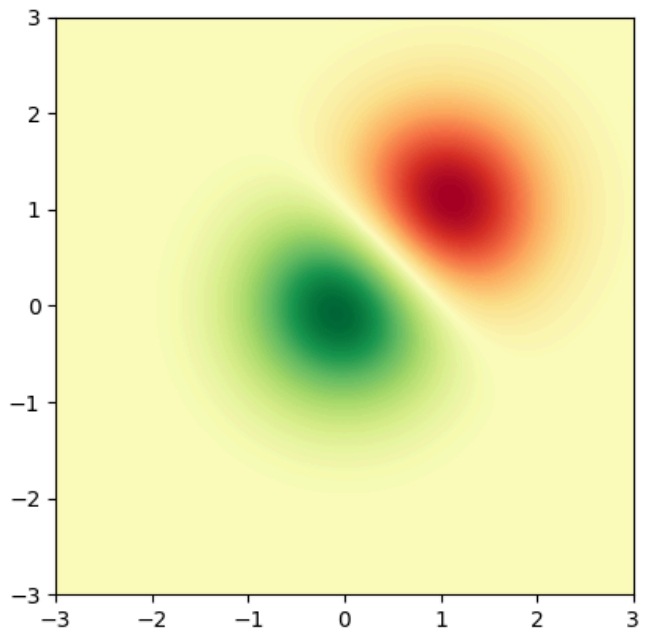
Many ways to plot images

The most common way to plot images in Matplotlib is with `imshow`. The following examples demonstrate much of the functionality of imshow and the many images you can create.

```
import matplotlib.pyplot as plt
import numpy as np

import matplotlib.cbook as cbook
import matplotlib.cm as cm
from matplotlib.patches import PathPatch
from matplotlib.path import Path

# Fixing random state for reproducibility
np.random.seed(19680801)
```

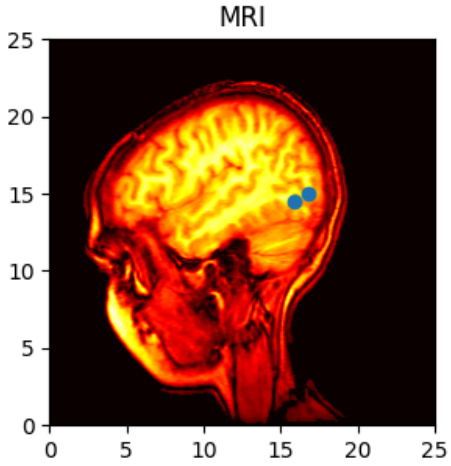
First we'll generate a simple bivariate normal distribution.

```
delta = 0.025
x = y = np.arange(-3.0, 3.0, delta)
X, Y = np.meshgrid(x, y)
Z1 = np.exp(-X**2 - Y**2)
Z2 = np.exp(-(X - 1)**2 - (Y - 1)**2)
Z = (Z1 - Z2) * 2

fig, ax = plt.subplots()
im = ax.imshow(Z, interpolation='bilinear', cmap=cm.RdYlGn,
               origin='lower', extent=[-3, 3, -3, 3],
               vmax=abs(Z).max(), vmin=-abs(Z).max())

plt.show()
```



It is also possible to show images of pictures.

```
# A sample image
with cbook.get_sample_data('grace_hopper.jpg') as image_file:
    image = plt.imread(image_file)

# And another image, using 256x256 16-bit integers.
w, h = 256, 256
with cbook.get_sample_data('s1045.ima.gz') as datafile:
    s = datafile.read()
A = np.frombuffer(s, np.uint16).astype(float).reshape((w, h))
extent = (0, 25, 0, 25)

fig, ax = plt.subplot_mosaic([
    ['hopper', 'mri']
], figsize=(7, 3.5))

ax['hopper'].imshow(image)
ax['hopper'].axis('off')  # clear x-axis and y-axis

im = ax['mri'].imshow(A, cmap=plt.cm.hot, origin='upper', extent=extent)

markers = [(15.9, 14.5), (16.8, 15)]
x, y = zip(*markers)
ax['mri'].plot(x, y, 'o')

ax['mri'].set_title('MRI')

plt.show()
```



Interpolating images

It is also possible to interpolate images before displaying them. Be careful, as this may manipulate the way your data looks, but it can be helpful for achieving the look you want. Below we'll display the same (small) array, interpolated with three different interpolation methods.

The center of the pixel at A[i, j] is plotted at (i+0.5, i+0.5). If you are using interpolation='nearest', the region bounded by (i, j) and (i+1, j+1) will have the same color. If you are using interpolation, the pixel center will have the same color as it does with nearest, but other pixels will be interpolated between the neighboring pixels.

To prevent edge effects when doing interpolation, Matplotlib pads the input array with identical pixels around the edge: if you have a 5x5 array with colors a-y as below:

```
a b c d e
f g h i j
k l m n o
p q r s t
u v w x y
```

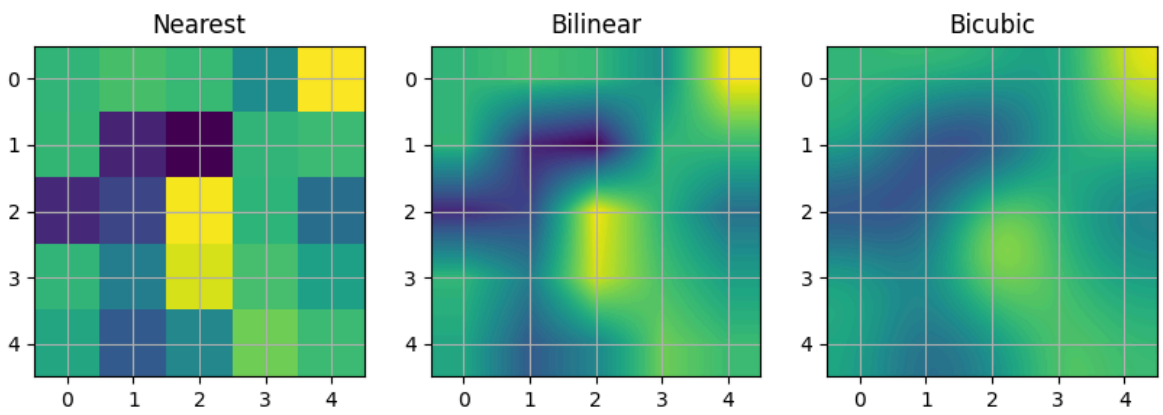
Matplotlib computes the interpolation and resizing on the padded array

```
a a b c d e e
a a b c d e e
f f g h i j j
k k l m n o o
p p q r s t t
o u v w x y y
o u v w x y y
```

and then extracts the central region of the result. (Extremely old versions of Matplotlib (<0.63) did not pad the array, but instead adjusted the view limits to hide the affected edge areas.)

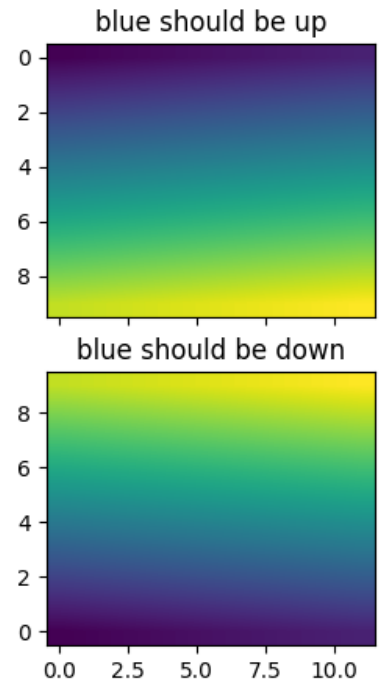
This approach allows plotting the full extent of an array without edge effects, and for example to layer multiple images of different sizes over one another with different interpolation methods -- see Layer Images. It also implies a performance hit, as this new temporary, padded array must be created. Sophisticated interpolation also implies a performance hit; for maximal performance or very large images, interpolation='nearest' is suggested.


```
A = np.random.rand(5, 5)

fig, axs = plt.subplots(1, 3, figsize=(10, 3))
for ax, interp in zip(axs, ['nearest', 'bilinear', 'bicubic']):
    ax.imshow(A, interpolation=interp)
    ax.set_title(interp.capitalize())
    ax.grid(True)

plt.show()
```



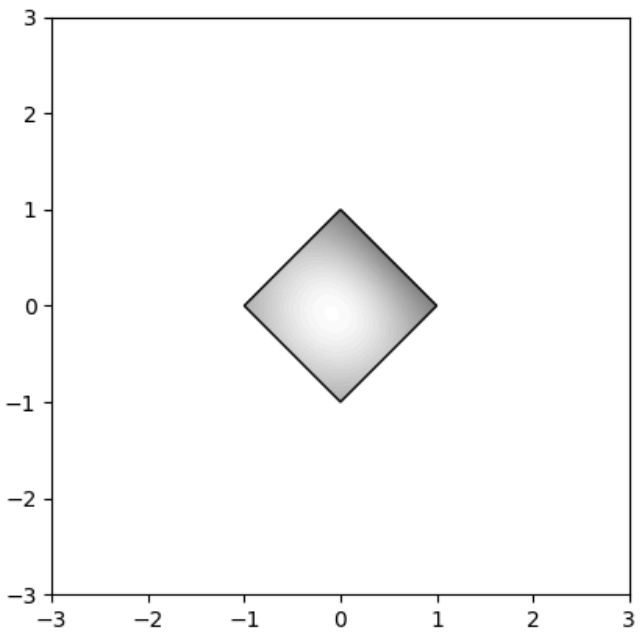
You can specify whether images should be plotted with the array origin x[0, 0] in the upper left or lower right by using the origin parameter. You can also control the default setting image.origin in your matplotlibrc file. For more on this topic see the complete guide on origin and extent.

```
x = np.arange(120).reshape((10, 12))

interp = 'bilinear'
fig, axs = plt.subplots(nrows=2, sharex=True, figsize=(3, 5))
axs[0].set_title('blue should be up')
axs[0].imshow(x, origin='upper', interpolation=interp)

axs[1].set_title('blue should be down')
axs[1].imshow(x, origin='lower', interpolation=interp)
plt.show()
```



Finally, we'll show an image using a clip path.

```
delta = 0.025
x = y = np.arange(-3.0, 3.0, delta)
X, Y = np.meshgrid(x, y)
Z1 = np.exp(-X**2 - Y**2)
Z2 = np.exp(-(X - 1)**2 - (Y - 1)**2)
Z = (Z1 - Z2) * 2

path = Path([[0, 1], [1, 0], [0, -1], [-1, 0], [0, 1]])
patch = PathPatch(path, facecolor='none')

fig, ax = plt.subplots()
ax.add_patch(patch)

im = ax.imshow(Z, interpolation='bilinear', cmap=cm.gray,
               origin='lower', extent=[-3, 3, -3, 3],
               clip_path=patch, clip_on=True)
im.set_clip_path(patch)

plt.show()
```



References

The use of the following functions, methods, classes and modules is shown in this example:

- `matplotlib.axes.Axes.imshow` / `matplotlib.pyplot.imshow`
- `matplotlib.artist.Artist.set_clip_path`
- `matplotlib.patches.PathPatch`

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